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Age and Gender Leucocytes Variances and References Values Generated Using the Standardized ONE-Study Protocol

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• **Abstract**

Flow cytometry is now accepted as an ideal technology to reveal changes in immune cell composition and function. However, it is also an error-prone and variable technology, which makes it difficult to reproduce findings across laboratories. We have recently developed a strategy to standardize whole blood flow cytometry. The performance of our protocols was challenged here by profiling samples from healthy volunteers to reveal age- and gender-dependent differences and to establish a standardized reference cohort for use in clinical trials. Whole blood samples from two different cohorts were analyzed (first cohort: $n = 52$, second cohort: $n = 46$, both 20–84 years with equal gender distribution). The second cohort was run as a validation cohort by a different operator. The "ONE Study" panels were applied to analyze expression of >30 different surface markers to enumerate proportional and absolute numbers of >50 leucocyte subsets. Indeed, analysis of the first cohort revealed significant age-dependent changes in subsets e.g. increased activated and differentiated $CD4^+$ and $CD8^+$ T cell subsets, acquisition of a memory phenotype for Tregs as well as decreased MDC2 and Marginal Zone B cells. Males and females showed different dynamics in age-dependent T cell activation and differentiation, indicating faster immunosenescence in males. Importantly, although both cohorts consisted of a small sample size, our standardized approach enabled validation of age-dependent changes with the second cohort. Thus, we have proven the utility of our strategy and generated reproducible reference ranges accounting for age- and gender-dependent differences, which are crucial for a better patient monitoring and individualized therapy.

• **Key terms**

immunosenescence; standardization; adaptive immunity; innate immunity; aging

THE immune system is very heterogeneous with regard to its composition and function. This has become even more evident upon recent development and application of multidimensional assays such as flow cytometry in clinical settings. Flow cytometry can and is being used for a wide range of diagnostics and management of diseases, immunotherapy and cancers (1–3). Immunological heterogeneity is however not only related to disease and medications but also to many individual factors such as age, gender, and even circadian rhythms (4,5).

The age-dependent deterioration of immunological functions is commonly referred to as immunosenescence (6). The declining function is associated with an increased mortality and morbidity by common infections such as pneumonia and influenza (7), and antibody-mediated response to vaccinations is impaired whereas the risk to develop malignant cancer is increased (8–10). In addition to immunosenescence, an increase of several proinflammatory cytokines such as IL-6 and TNF-α is observed in the elderly (11). This opposing process is termed inflammaging and is

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Table 1. Overview of age and gender distribution in study population

COHORT 1	MALE		FEMALE		TOTAL	
AGE	*N*	MEAN AGE	*N*	MEAN AGE	*N*	MEAN AGE
20-29	5	25.2	5	26.0	10	26.6
30-39	6	31.5	6	33.7	12	32.6
40-49	5	42.6	5	44.4	10	43.5
50-59	5	54.8	5	55.0	10	54.9
60-75	5	65.2	5	67.2	10	66.2
Total	26	43.4	26	44.8	52	44.1

COHORT 2	MALE		FEMALE		TOTAL	
AGE GROUP	*N*	MEAN AGE	*N*	MEAN AGE	*N*	MEAN AGE
20-29	5	25.6	6	25.0	11	25.3
30-39	5	33.4	5	34.2	10	33.8
40-49	4	42.0	4	44.3	8	43.1
50-59	4	54.8	5	55.0	9	54.9
60-84	4	70.5	4	63.8	8	67.1
Total	22	42.4	24	44.4	46	43.4

characterized as a chronic state of unspecific low-grade inflammatory activity linked to the occurrence of autoimmune disease and atherosclerosis (12,13)

Several studies have reported age-dependent cellular changes to the immune system, for example, changes in number and function of innate immune cells subsets as well as subsets of the adaptive immune system (4,14–22). Age associated changes are most pronounced for the adaptive immune cell subsets.

Gender also plays a role in immunological heterogeneity. The most obvious evidence is the clinical manifestations of several autoimmune diseases such as Sjögren's disease or systemic lupus erythematosus that have a strong female:male bias (23,24). It is commonly believed that gender-related differences result primarily from differences in sex hormones, which is also supported by the fact that hormonal irregularities (e.g., during pregnancy) can lead to flares and/or amelioration of autoimmune diseases (25–27).

While aging and gender effects on immune cell composition and function have been thoroughly studied, most studies analyzed only narrow aspects of the immune system, for example, only T cells or NK cells, and sort participants into differently defined "young" and "old" categories. With both innate and adaptive immune cell subsets being simultaneously altered, and age being a continuous factor, it becomes obvious that a more global and multi-parametric approach is warranted.

Conflicting results regarding age-dependent subset changes have been reported (28) indicating that immune cell aging is also influenced by factors such as ethnical or environmental variables, requiring methods to standardize and harmonize immune cell profiling (29–32). Indeed, multiparametric flow cytometry is a very complex method, which in combination with a diversity of available equipment, reagents and preanalytical factors such as specimen age, and staining procedures, compensation but also analytical factors such as subset definition increases the variability, particularly when comparing results obtained within different laboratories. All these variables need to be controlled to ensure standardization. We could recently show that this is potentially possible also in a multicenter clinical trial setting (Streitz et al., 2013).

Moreover, the establishment of age- and gender-based standardized reference values would improve the diagnostics necessary for a more individualized treatment of inflammatory diseases.

In this study, we tested the performance of our recently established and standardized set of six flow "ONE Study" panels capturing over 50 different leucocyte subsets to reproducibly reveal age- and gender-dependent changes in whole blood samples (33). We studied samples from healthy individuals to verify previously described age effects and to identify and validate new influences of age and gender on leucocyte composition. We show that specific subpopulations of the adaptive as well as innate immune system change continuously during aging. Although, increases in differentiated/senescent T cell subsets dominate age-dependent alterations, we also detected changes in B cell subsets such as Marginal Zone (MZ) B cells. Interestingly, T cell aging seems to occur more slowly in females compared with males. In addition, we validated our results in a second participant cohort performed by a different operator. We thus sought to achieve both a validation of our finding both regarding the reference leucocyte blood counts and of the standardization of our method.

Methods

Study Population

The studied individuals were adults and divided into two cohorts. The first cohort consisted of samples from 52 individuals ranging from 20 to 75 years of age with an equal age and gender distribution (Table 1). The second, validation cohort consisted of samples from 46 individuals ranging from 20 to 84 years and equivalent gender distribution. The participants were mainly found among the hospital staff (Charité Universitätsmedizin) and belonged to European and Caucasian ethnicities. They were defined as healthy when they were not excluded by the following criteria: Symptomatic infections within the previous two weeks including fever, immunizations within the last month, pregnancy, inflammatory disease, or the use of medication with suspected effects on the immune system, for example, antihistamines. The participants gave their written and informed consent prior to participating in the study. The study was approved by local ethics committee (Ethikkommision der Charité). The CMV status or existence other of chronic viral infections were not assessed in this study.

Sample Preparation and Flow Cytometry

Whole blood staining and subsequent flow cytometric analysis was performed according to the protocol developed by the ONE-Study Consortium (33). A 2 mL EDTA-anticoagulated blood sample was collected by venipuncture for antibody staining and another 2 mL sample was sent to the local hospital laboratory (Labor Berlin) for a standard leucocyte cell count. All blood samples were collected before noon, stained within 4 h of collection and analyzed the same day by flow cytometry. Monoclonal antibodies conjugated to FITC, PE, ECD, PC7, APC, APC-A700, APC-A750, PacBlue, or KrOrange were used to stain a total of 30 different epitopes divided into 6 different panels (Supporting Information Table S1). All antibodies were obtained from Beckman Coulter (Marseille, France), except anti-BDCA-2 and anti-BDCA-3 from Miltenyi Biotec (Bergisch Gladbach, Germany) and anti-CCR7 from R&D Systems (Wiesbaden, Germany). In panel 1–4, 100 µL whole blood was incubated with panel antibodies for 15 min at room temperature and the erythrocytes were then lysed for 15 min with a lyse-fix solution consisting of Versa Lyse™ and IOTest® Fixative Solution (Beckman Coulter). To achieve sufficient cell numbers in panel 6, two samples of 100 µL whole blood were stained and combined prior to acquisition. 300 µL of blood was used in panel 5, the erythrocytes were lysed for 12 min with Red Blood Cell Lysis Solution (Miltenyi) before the antibody staining and the sample was incubated for 20 min at 4°C. All samples were analyzed on a 10-Color Navios Flow Cytometer (Beckman Coulter) after a daily calibration with Flow-Set Pro Beads (Beckman Coulter). Samples were acquired at fixed measuring time intervals to ensure acquisition of the maximum volume available (~85% of initial blood volume).

Absolute cell numbers were determined by relating the $CD45^+$ count from the flow cytometry to the total leucocyte count. To assure optimal calibration, a daily verification of optical alignment and fluidic systems with “Flow Check Pro” and setup performance check with “Flow-Set Pro Beads” (Beckman Coulter) was performed. Both cohorts were stained identically with the exception of one cell type Tregs. Although these were defined as $CD4^+$ $CD25^{high}$ $CD127^{low}$ in both the first and second cohort, the respective Treg phenotype has been enumerated in the second cohort by an intracellular ($FoxP3^+$) staining protocol which included CD25 and CD127.

Data Analysis and Statistics

The flow cytometry data analysis was performed in Kaluza version 1.2 (Beckman Coulter) with a standardized gating strategy (Table 2) performed by a single operator according to the ONE-Study protocol. Leucocytes were identified by SSC and FSC and doublets eliminated by measuring time of flight (TOF). R, version 3.1 was used for the statistical analysis of gated populations as well as for visualizing the data in scatter plots and heatmaps (34). Spearman correlation matrices served as input for hierarchical clustering shown in heatmaps of all leucocyte proportions and absolute counts. Since most cellular subsets are non-normally distributed and exhibit a high degree of heteroscedasticity, significant association with age was examined by Spearman correlation coefficients. In addition, linear regression models were generated by robust MM-estimation to provide a reasonable comparison and to illustrate the continuous effect of age (35).

Gender specific differences in subset proportions and blood counts were tested with the Wilcoxon-Mann-Whitney rank-sum test. A heteroscedastic global test was applied to the robust linear regression estimates to assess a gender-specific association of subsets with age (36,37). For individual subsets this was extended to a robust ANCOVA to identify differences within each age decade (38). Equivalence of subset norm-means established in the first cohort and subset means validation in the second cohort was assessed by a robust two one-sided test (TOST) using Yuen’s t test of trimmed means and winsorized variances (39–41). A significant result was defined as $P < 0.05$ after adjusting for multiple comparisons by controlling the false discovery rate at the same nominal level (42).

Results

Change of Innate and Adaptive Leucocyte Subsets with Age

In order to test the performance of our previously described protocols for whole blood flow cytometry, we studied changes to 56 leucocyte subsets or parameters according to age in a cohort of 52 healthy individuals spanning an age differences of over 50 years with an equal gender distribution (Cohort 1). The absolute and proportional medians with minimum and maximum values for the total cohort population are presented in Table 2. The proportional medians with upper and lower quartiles of the different age decades and the p-values from an age-dependent statistical analysis are presented in Table 3 and the absolute medians with interquartile ranges and P-values in Supporting Information Table S2.

Figure 1A shows hierarchical clustering of all subsets’ proportional data across all samples analyzed in a heatmap representation. This allowed us to identify groups of individuals with similar leucocyte subset compositions, and it appears that samples segregate into two main clusters, one containing mostly older participants above 40 years of age (right cluster) and the other mostly younger participants below 40. Especially activated, for example, $HLA\text{-}DR^+$ and differentiated, for

Table 2. The description and phenotypes of all investigated cell subsets and the medians with minimum and maximum of both proportional and absolute data in Cohort 1

ALL CELL SUBSETS FIRST COHORT			MEDIAN (RANGE)	
DESCRIPTION	PHENOTYPE	DENOMINATOR	PROPORTIONAL (%)	ABSOLUTE (CELLS/μL)
Leucocytes	$CD45^+$	–	–	5610 (3770–9850)
Granulocytes	$CD45^+$ SS (high)	Leucocytes	58.5 (37.2–80.5)	3094 (1526–7306)
All monocytes	$CD14^+$	Leucocytes	7.1 (3–10.9)	426 (144–702)
Classical monocytes	$CD14^{high}$ $CD16^-$	Monocytes	85.7 (68.4–93.4)	382 (114–589)
Intermediate monocytes	$CD14^{high}$ $CD16^+$	Monocytes	5.8 (2.6–15.8)	24 (7–70)
Nonclassical monocytes	$CD14^+$ $CD16^{high}$	Monocytes	6 (2.2–16.7)	23 (7–86)
Dendritic cells	Lin^- $HLA\text{-}DR^+$	Leucocytes	0.8 (0.2–1.9)	42 (20–121)
Plasmacytoid dendritic cells	Lin^- $HLA\text{-}DR^+$ $CD11c^-$ $CD123^+$ $BDCA2^+$	Leucocytes	0.1 (0–0.4)	8 (1–21)
Myeloid dendritic cells	Lin^- $HLA\text{-}DR^+$ $CD11c^+$	Leucocytes	0.7 (0.1–1.7)	37 (10–107)
$CD16^+$ mDCs	Lin^- $HLA\text{-}DR^+$ $CD11c^+$ $CD16^+$ $BDCA3^-$	mDCs	75.9 (33.9–98.2)	30 (5–95)
MDC1	Lin^- $HLA\text{-}DR^+$ $CD11c^+$ CD16- $BDCA3^-$	mDCs	23.2 (1.7–61.6)	9 (1–22)
MDC2	Lin^- $HLA\text{-}DR^+$ $CD11c^+$ CD16- $BDCA3^+$	mDCs	0.9 (0.1–4.5)	0.3 (0–0.9)
Lymphocytes	$CD45^+$ SS (low) FS (low)	Leucocytes	32.6 (11.4–57)	1875 (435–3389)
NK-cells	$CD3^-$ $CD56^+$	Lymphocytes	13.2 (3.3–32.9)	230 (53–569)
Mature NK-cells	$CD3^-$ $CD56^{dim}$	NK-cells	94.9 (71.9–99.5)	213 (48–557)
Immature NK-cells	$CD3^-$ $CD56^{high}$	NK-cells	5.1 (0.5–28.1)	11 (2–33)
T cells	$CD3^+$	Lymphocytes	69.5 (53.7–82.8)	1342 (270–2586)
$CD4^+$ cells	$CD3^+$ $CD4^+$	T cells	61 (46.2–78)	799 (199–1414)
Activated $CD4^+$ T cells	$CD3^+$ $CD4^+$ $HLA\text{-}DR^+$	$CD4^+$ cells	8 (3.6–31.4)	64 (18–223)
Activated $CD4^+$ T cells	$CD3^+$ $CD4^+$ $HLA\text{-}DR^+$ $CD45RA^+$	$CD4^+$ cells	0.3 (0.1–3.9)	2 (0–39)
Activated memory $CD4^+$ T cells	$CD3^+$ $CD4^+$ $HLA\text{-}DR^+$ $CD45RA^-$	$CD4^+$ cells	7.5 (3–30.4)	61 (14–205)
Naïve $CD4^+$ T cells	$CD3^+$ $CD4^+$ $CCR7^+$ $CD45^+$	$CD4^+$ cells	36.1 (7.2–68.9)	289 (45–1079)
Central memory $CD4^+$ T cells	$CD3^+$ $CD4^+$ $CCR7^+$ CD45-	$CD4^+$ cells	39.5 (15–64.3)	317 (70–671)
Effector memory $CD4^+$ T cells	$CD3^+$ $CD4^+$ $CCR7^-$ $CD45^-$	$CD4^+$ cells	13.9 (4.9–43.6)	119 (23–301)
Effector memory $CD45RA^+$ $CD4^+$ T cells	$CD3^+$ $CD4^+$ $CCR7^-$ CD45+	$CD4^+$ cells	0.4 (0.1–11.7)	4 (0–141)
Differentiated/senescent $CD4^+$ T cells	$CD3^+$ $CD4^+$ $CD27^-$	$CD4^+$ cells	6.5 (1.5–36.9)	52 (14–320)
Differentiated/senescent $CD4^+$ T cells	$CD3^+$ $CD4^+$ $CD28^-$	$CD4^+$ cells	0.8 (0–26.2)	6 (0–174)
Differentiated/senescent $CD4^+$ T cells	$CD3^+$ $CD4^+$ $CD57^+$	$CD4^+$ cells	1.9 (0.6–30.4)	15 (3–191)
Regulatory T cells (Tregs)	$CD3^+$ $CD4^+$ $CD25^{high}$ $CD127^{low}$	$CD4^+$ cells	7.9 (5.1–12.7)	66 (28–142)
Naïve Tregs	$CD3^+$ $CD4^+$ $CD25^{high}$ $CD127^{low}$ $CD45RA^+$	$CD4^+$ cells	24.8 (3.5–77.3)	13 (4–68)
Memory Tregs	$CD3^+$ $CD4^+$ $CD25^{high}$ $CD127^{low}$ $CD45RA^-$	$CD4^+$ cells	75.2 (22.7–96.5)	51 (6–118)
$CD8^+$ T cells	$CD3^+$ $CD8^+$	T cells	32.5 (14.8–48.4)	438 (61–1118)
Activated $CD8^+$ T cells	$CD3^+$ $CD8^+$ $HLA\text{-}DR^+$	CD8+ T cells	29.9 (6.1–63.5)	120 (22–689)
Activated $CD8^+$ T cells	$CD3^+$ $CD8^+$ $HLA\text{-}DR^+$ $CD45RA^+$	CD8+ T cells	11.8 (1–43.4)	38 (5–458)

TABLE 2. *Continued*

ALL CELL SUBSETS FIRST COHORT			MEDIAN (RANGE)	
DESCRIPTION	PHENOTYPE	DENOMINATOR	PROPORTIONAL (%)	ABSOLUTE (CELLS/μL)
Activated memory CD8$^+$ T cells	CD3$^+$ CD8$^+$ HLA-DR$^+$ CD45RA$^-$	CD8+ T cells	16.4 (3.7–53.3)	71 (13–311)
Naïve CD8$^+$ T cells	CD3$^+$ CD8$^+$ CCR7$^+$ CD45$^+$	CD8+ T cells	26.2 (2.6–72.4)	111 (18–355)
Central Memory CD8$^+$ T cells	CD3$^+$ CD8$^+$ CCR7$^+$ CD45$^-$	CD8+ T cells	11.7 (2.7–36.2)	48 (7–206)
Effector memory CD8$^+$ T cells	CD3$^+$ CD8$^+$ CCR7$^-$ CD45$^-$	CD8+ T cells	34.1 (4.7–60.1)	155 (13–457)
Effector memory CD45RA$^+$ CD8$^+$ T cells	CD3$^+$ CD8$^+$ CCR7$^-$ CD45$^+$	CD8+ T cells	18.6 (1.6–62)	62 (8–635)
Differentiated/senescent CD8$^+$ T cells	CD3$^+$ CD8$^+$ CD27$^-$	CD8+ T cells	21.6 (2.5–60.2)	78 (11–666)
Differentiated/senescent CD8$^+$ T cells	CD3$^+$ CD8$^+$ CD28$^-$	CD8+ T cells	27.9 (4.4–68.8)	102 (18–760)
Differentiated/senescent CD8$^+$ T cells	CD3$^+$ CD8$^+$ CD57$^+$	CD8+ T cells	24.6 (4.6–59.6)	98 (17–600)
TCR-gd T cells	TCR-gd CD3$^+$	T cells	3.8 (0.4–20.7)	49 (5–334)
TCR-gd CD4$^+$ T cells	TCR-gd CD3$^+$ CD4+	TCR-gd T cells	0.4 (0–7.2)	0.2 (0–3.3)
TCR-gd CD8$^+$ T cells	TCR-gd CD3$^+$ CD8+	TCR-gd T cells	14.6 (3–57.1)	8 (0–50)
TCR-gd CD4$^-$ CD8- T cells	TCR-gd CD3$^+$ CD4- CD8-	TCR-gd T cells	84.7 (42.8–96.4)	38 (4–307)
CD4$^+$:CD8$^+$ ratio	CD3$^+$ CD4$^+$: CD3$^+$ CD8$^+$	–	–	1.9 (1–4.9)
B cells	CD19$^+$ CD3$^-$	Lymphocytes	11.3 (3.8–21.5)	205 (51–728)
Naïve B cells	CD19$^+$ CD27$^-$ IgD$^+$	B cells	53.6 (17.9–75.1)	43 (5–401)
Transitional B cells	CD19$^+$ CD27$^-$ CD38^{high} IgM$^+$ CD24$^+$	B cells	2.1 (0.1–6.3)	2 (0–20)
MZ B cells	CD19$^+$ CD27$^+$ IgD$^+$	B cells	12.5 (3.8–35.6)	8 (1–84)
Memory B cells	CD19$^+$ CD27$^+$ CD38^{dim}	B cells	21 (11–46.6)	18 (3–80)
Class-switched B cells	CD19$^+$ CD27$^+$ CD38^{dim} IgD$^-$ IgM$^-$	B cells	15.9 (4.6–35.5)	14 (1–53)
Non-switched B cells	CD19$^+$ CD27$^+$ CD38^{dim} IgM$^+$	B cells	6 (1.9–23.7)	5 (1–28)
Plasmablasts	CD19$^+$ CD27^{high} CD38^{high} IgD$^-$ IgM$^-$	B cells	1.2 (0.3–7.8)	1 (0–6)
CD21low B cells	CD19$^+$ CD38^{low} CD21^{low}	B cells	6.2 (1.9–19.3)	5 (1–37)

example, CD57$^+$ and CD27$^-$ T cell subsets contribute to this separation with higher frequencies (marked in red) observed in samples from elderly individuals, whereas samples from younger individuals contain fewer proportions. Interestingly, one cluster with very low activated and differentiated T cell subsets (dark blue) is made up preferentially by samples from young female individuals. In contrast, samples from young males are more evenly distributed between clusters with low, intermediate, or high activated/differentiated CD8$^+$ and/or CD4$^+$ T cell subsets. This indicates a slower "T cell aging" in females. Generating a heatmap for the absolute data (Fig. 1B) also results in separation of groups mainly according to age. However, the generated clusters are different, and the separation is dominated by one group displaying very high absolute values for the above-mentioned T cell subsets.

The overall leucocyte count showed no significant change with age and had a median of 5610 cells/μL in this cohort. Numbers of cells of the innate immune system also remained largely unchanged. Monocyte subsets, especially nonclassical CD14$^+$ CD16^{high} monocytes showed a tendency toward an age-dependent increase in absolute numbers over five decades (Supporting Information Table S2). The only cells of the innate immune system that reliably correlated with age were myeloid dendritic cells (mDCs) and CD16$^+$ mDCs increasing from 33 and 26 cells/μL in the youngest, to 65 and 52 cells/μL, respectively, in the oldest participants.

The number of NK cells and their respective subsets did not change significantly with age, either proportionally or absolutely (Table 3 and Supporting Information Table S2). The median of all NK cells was 13.2% of total lymphocytes with the majority being the CD56^{dim} (94.9%) and only 5.1% CD56^{high}.

The number of total T cells also did not change significantly with age and we found a median of 69.5% of all

Table 3. The medians with interquartile range of all cell subset proportions (%) divided into 5 age groups

PROPORTIONAL NUMBERS, FIRST COHORT	MEDIAN (INTERQUARTILE RANGE)					CORRELATION	
CELL SUBSET	20–30	30–40	40–50	50–60	60–75	RHO	*P*
$CD45^{+}$ Leucocytes	100 (100–100)	100 (100–100)	100 (100–100)	100 (100–100)	100 (100–100)		
Granulocytes	55.8 (45.6–62.9)	52.8 (45.4–62.9)	60.2 (55–63.6)	59.5 (55.2–61.8)	58.6 (50.9–67)		
$CD14^{+}$ Monocytes	8.1 (5.2–10.5)	6 (5.4–7.3)	7 (5.4–8)	6.6 (5.1–8.7)	7.9 (7.1–8.5)		
$CD14^{high}$ $CD16^{-}$	83.3 (79.2–88.2)	89.5 (80.4–91.9)	89.3 (83–91.1)	84.7 (82.7–88.7)	84.5 (83–87.5)		
$CD14^{high}$ $CD16^{+}$	7.9 (4.9–11.4)	4.4 (3.5–9.6)	4.8 (3.5–8.2)	6.1 (5.1–7.2)	7.1 (5–8.1)		
$CD16^{high}$	5.9 (5–7)	4.3 (3–7.1)	4.9 (3.9–8.3)	7.8 (4.7–8.9)	6.7 (5.5–7.6)		
Dendritic Cells	0.9 (0.7–1.2)	0.6 (0.5–0.8)	0.9 (0.6–1)	0.9 (0.4–1.1)	1.1 (0.7–1.4)		
pDCs	0.2 (0.1–0.3)	0.1 (0–0.2)	0.1 (0.1–0.2)	0.1 (0.1–0.2)	0.1 (0.1–0.2)		
mDCs	0.7 (0.5–0.9)	0.5 (0.3–0.7)	0.7 (0.4–0.8)	0.8 (0.3–0.9)	1 (0.6–1.3)		
$CD16^{+}$ mDCs	68.2 (60.5–80.6)	61.8 (51.2–90.8)	77.7 (74.1–86.4)	75.1 (60.5–80.9)	77 (74.7–83.9)		
MDC1	30.3 (18.3–37.3)	37.1 (8.8–47.6)	20.9 (12.9–24.4)	24.7 (18.3–38.5)	22.1 (15.5–24.5)		
MDC2	**1.3 (0.9–2)**	**0.9 (0.3–1.8)**	**1.1 (0.6–1.5)**	**0.7 (0.4–1.7)**	**0.8 (0.5–0.9)**	**–0.35**	**0.033**
Lymphocytes	33.6 (24–46.2)	39.8 (29.4–46.4)	30.5 (27.3–38.2)	32.1 (30.3–35.5)	32.7 (24.4–39.2)		
$CD56^{+}$ NK cells	13.6 (8.7–16.1)	11 (8.4–13)	14.1 (11.7–16.2)	14.1 (7.9–22.3)	13.6 (7.8–18.7)		
$CD56^{dim}$	94.2 (89.4–96.9)	94.6 (92.4–95.6)	95.5 (94.3–97.3)	96.1 (92.2–98.3)	95.8 (93.6–96.9)		
$CD56^{high}$	5.8 (3.2–10.7)	5.4 (4.4–7.6)	4.5 (2.7–5.7)	3.9 (1.7–7.8)	4.2 (3.1–6.4)		
$CD3^{+}$ T cells	70 (64.9–75.7)	74.7 (68.3–76.7)	68.1 (66.6–70.3)	68.6 (62.1–75.8)	69.2 (66.1–77)		
$CD4^{+}$	59.4 (56.7–68.6)	61.9 (53.1–69)	61.7 (50.9–66.4)	60.8 (51.1–66.3)	61.2 (54.4–63.9)		
HLA-DR^{+}	**5.4 (4.4–7.4)**	**7.1 (4.1–12.7)**	**7.9 (4.9–11.9)**	**13.4 (10.7–14)**	**10.9 (6.7–13.2)**	**0.37^{co}**	**0.025**
$CD45RA^{+}$	0.2 (0.1–0.3)	0.3 (0.2–1.8)	0.2 (0.2–0.7)	0.5 (0.2–0.9)	0.2 (0.2–1)		
CD45RA^{-}	**5.3 (4.1–7.2)**	**6.6 (3.8–10.2)**	**7.5 (4.7–10.1)**	**12.9 (10–13.5)**	**10.7 (6.6–12.4)**	**0.39^{co}**	**0.016**
Naive	**45 (36.4–48)**	**41.5 (39–53.7)**	**32.9 (25.5–42.6)**	**21.3 (16.2–36.7)**	**20.2 (16.2–31.2)**	**–0.57^{co}**	**<0.001**
CM	**36.2 (31.6–39.6)**	**33.8 (30.7–40.4)**	**43 (31.9–45.3)**	**44.2 (38.1—51.3)**	**47.8 (43.5–53.7)**	**0.52^{co}**	**0.001**
EM	**13.7 (10.7–15.1)**	**12.4 (7.2–15.3)**	**17 (11.4–20.4)**	**19 (11.8–26.5)**	**15.5 (12.2–23.8)**	**0.36**	**0.029**
TEMRA	0.4 (0.2–0.5)	0.2 (0.2–2.8)	0.3 (0.3–1.1)	1 (0.2–2.2)	0.4 (0.2–4.5)		
CD27^{-}	**4.4 (4.2–5.9)**	**5.3 (3.7–10.9)**	**7.3 (3.6–14.2)**	**12.7 (5.8–19.3)**	**7.5 (6.1–21.7)**	**0.46**	**0.005**
$CD28^{-}$	0.2 (0.1–0.5)	0.2 (0.1–2.9)	0.9 (0.1–3.4)	2.7 (0.9–7.7)	1.1 (0.3–8)		
CD57^{+}	**1 (0.8–1.6)**	**1.7 (0.8–4.2)**	**2.2 (1.1–7)**	**5.2 (1.2–9.1)**	**1.9 (1.4–11)**	**0.37**	**0.028**
Tregs	7.8 (6.9–8.9)	8 (6.4–10.5)	8.2 (6.9–9.1)	7.6 (7.2–8.8)	8.6 (7.8–9.7)		
Naive	**31.7 (26.7–36.7)**	**30.6 (26.9–40.8)**	**26.5 (17.9–32.6)**	**11.9 (9.6–15.7)**	**10.2 (8.7–15.6)**	**–0.72^{na}**	**<0.001**
Memory	**68.3 (63.3–73.3)**	**69.4 (59.2–73.1)**	**73.5 (67.5–82.1)**	**88.1 (84.3–90.4)**	**89.8 (84.4–91.3)**	**0.72^{na}**	**<0.001**
$CD8^{+}$	32.2 (25.4–35.5)	32 (25.6–39.4)	33.4 (29.7–45.4)	31.9 (28.1–36.1)	33.6 (28–39.5)		
HLA-DR^{+}	**20.2 (15–37.6)**	**17.6 (9.1–44.6)**	**26.8 (13.2–37.3)**	**40.5 (28.5–53.5)**	**44.1 (34.2–49.8)**	**0.46^{co}**	**0.005**
HLA-DR^{+} CD45RA^{+}	**7.3 (3.5–9.6)**	**5.3 (2–24.1)**	**10.4 (3.1–17.5)**	**12.9 (6.2–18.4)**	**19.6 (10.8–29.7)**	**0.34^{co}**	**0.041**
HLA-DR^{+} CD45RA^{-}	**12.3 (9.3–20.7)**	**10 (6.7–15.7)**	**16.8 (10–18.6)**	**24.5 (18.3–32.2)**	**21 (12.7–29.3)**	**0.43^{co}**	**0.007**

TABLE 3. *Continued*

PROPORTIONAL NUMBERS, FIRST COHORT	MEDIAN (INTERQUARTILE RANGE)					CORRELATION	
CELL SUBSET	20–30	30–40	40–50	50–60	60–75	RHO	*P*
Naïve	**43.4 (24.1–53.2)**	**42.4 (29.2–65.9)**	**26.6 (23.5–38.2)**	**11.6 (7.8–24.5)**	**10.3 (5–18.7)**	**–0.71^{co}**	**<0.001**
CM	9.9 (7.5–15.1)	9.8 (6.2–14.1)	13.7 (11.3–17.3)	17.4 (10.2–22.6)	11.8 (9.5–16.2)		
EM	**34 (23.1–42)**	**23.3 (16.6–34.6)**	**34.4 (26.8–43.1)**	**43 (31.8–54.9)**	**37.6 (30.1–42.3)**	**0.35**	**0.035**
TEMRA	**11.5 (7.2–18.2)**	**10.1 (5.9–38.1)**	**16.6 (10.7–28.2)**	**22.5 (9.4–36.6)**	**42.4 (24.3–47.8)**	**0.44^{co}**	**0.006**
CD27$^-$	**8.9 (5.9–18.9)**	**6.2 (4.5–32.4)**	**22.4 (9–37.5)**	**28.6 (18.3–44.4)**	**39.7 (31.5–54.4)**	**0.57^{co}**	**<0.001**
CD28$^-$	**21.9 (14.3–27.7)**	**14 (7.7–43.3)**	**28.4 (15–41.8)**	**33.8 (20.5–50.8)**	**47.4 (37.7–56.4)**	**0.44^{co}**	**0.006**
CD57$^+$	**15.9 (12–25.1)**	**11.9 (9.8–35.3)**	**25.1 (16–42.6)**	**29 (19.2–47.1)**	**41.4 (30.9–50.3)**	**0.51^{co}**	**0.001**
TCR-gd	4.4 (1.9–8.7)	3.8 (2.9–5.5)	3.1 (1.9–3.9)	4.8 (2.2–10)	4.3 (2.3–7.4)		
CD4$^+$	**0.5 (0.3–2)**	**0.6 (0.2–1.3)**	**0.8 (0.2–1.1)**	**0.2 (0.1–1.1)**	**0.2 (0.1–0.4)**	**–0.41^{co}**	**0.012**
CD8$^+$	11.8 (6.7–17.8)	12.5 (10.4–23.2)	13.9 (10.2–25.1)	15.6 (12.4–22.4)	24 (14.6–42.4)		
CD4$^-$ CD8$^-$	87.1 (77.1–92.6)	86.7 (76.5–88.5)	85.8 (74.1–87.8)	83.5 (74.3–87.4)	75.7 (56.7–85.1)		
CD4$^+$:CD8$^+$	1.9 (1.7–2.6)	2 (1.3–2.6)	1.9 (1.1–2.2)	1.9 (1.4–2.2)	1.8 (1.4–2.3)		
CD19$^+$ B cells	10.8 (8.4–15.9)	12 (8.7–14.3)	12.3 (9.3–13.7)	10.6 (7.7–13.1)	11.5 (8.3–12.5)		
Naïve B cells	53 (46.4–61.8)	58.2 (38.4–62.9)	49 (35.9–62.6)	56.6 (37.7–70.9)	52.2 (37.8–69.2)		
Transitional B cells	1.5 (1.2–2.7)	2 (1–2.5)	2.5 (1.1–4.2)	3 (1.6–4.7)	2.8 (1.5–4.2)		
MZ B cells	**12.3 (10.7–14.6)**	**11.8 (9.8–18.9)**	**14.4 (12.8–20.1)**	**9.6 (5.6–17.1)**	**7.6 (4.7–12.3)**	**–0.32^{co}**	**ns**
Memory B cells	23.2 (15.9–27.3)	19.4 (17.1–22.7)	21.8 (15.9–35.4)	20.2 (16.1–35)	23.2 (15.3–30.6)		
Class-switched B cells	17.5 (8.5–21.4)	13.6 (11.6–17.9)	14.5 (9.6–21.4)	16.2 (10.7–23.2)	15.1 (9.2–23.6)		
Non-switched B cells	5.6 (3.6–7.3)	5.6 (4.5–8.5)	8.9 (5.4–12.2)	5.8 (4.5–8.7)	6 (3.5–10)		
Plasmablasts	1.6 (0.5–2.9)	1 (0.8–1.5)	1.2 (0.6–2.4)	1.4 (0.8–2.7)	1.4 (1.1–2.9)		
CD21^{low} B cells	5.2 (3.7–7.5)	5.5 (3.8–9.7)	7.8 (5.3–13.2)	8.4 (4.8–13.6)	6.9 (3.9–11.2)		

The third column highlights cell populations, which significantly correlate with age. Only Spearman correlation coefficients (rho) >0.3 or <−0.3 and respective *P*-values are reported.

[co]Significant correlation confirmed in second cohort ([na]subset not analysed). The subset denominator is listed in Table 2.

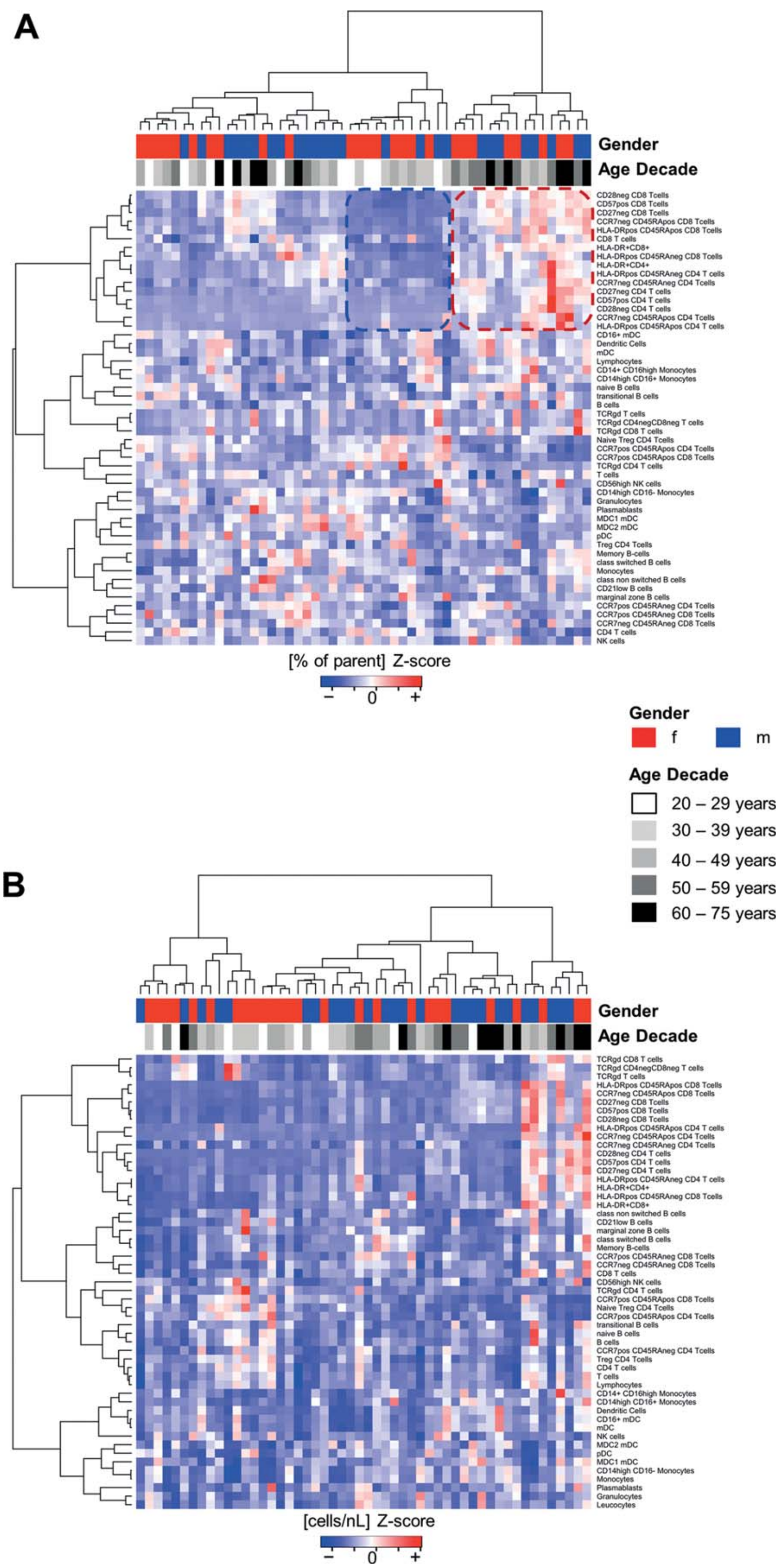

Figure 1. Cluster analysis of all proportional **(A)** and absolute **(B)** data acquired in the first cohort. The measured values of subset proportions, or absolute blood cell numbers, respectively, for each healthy individual are displayed in heatmaps, which are colored according to (per-subset) mean-centered and sigma-normalized data (Z-score). The height of the dendrogram represents the dissimilarity between clusters of healthy individuals (columns) or immune cell subsets (rows), respectively. The top-bars indicate the gender and age decade of each individual.

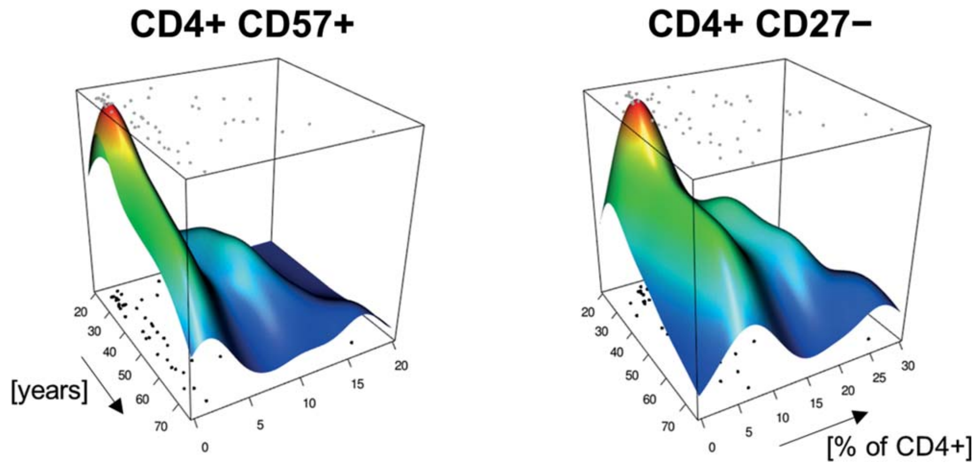

Figure 2. Segregated CD4$^+$ T cell senescence over age. The plot shows proportions of CD57$^+$ (left) and CD27$^-$ (right) among CD4$^+$ T cells over age on the x-y-plane, and in z-direction the respective kernel density estimate, illustrating the nonuniform distribution of individuals among the first cohort. For both subsets young individuals concentrate in a single high (red) density peak, which segregates with advancing age into two ridges, one with individuals remaining at low frequencies of activated T cells and the second indicating a distinct group of individuals with ever increasing frequencies of CD4$^+$ CD57$^+$ and CD4$^+$ CD27$^-$ T cells.

lymphocytes to be CD3$^+$ T cells. Similarly, the distribution of CD4$^+$ and CD8$^+$ T cells remained stable with the ratio of 1.9. In contrast, the distribution of naïve and memory T cells changed with age. Naïve CD4$^+$ and CD8$^+$ T cells were reduced from proportions of 45% and 43.3% to 20.2% ($P < 0.001$) and 10.3% ($P < 0.001$), respectively (Table 2). Conversely, we detected an increase in memory T cells such as CD45RA$^-$CCR7$^+$ central memory CD4$^+$ T cells from 36.2 to 47.8% ($P = 0.001$) and CD45RA$^+$ CCR7$^-$ terminal differentiated effector memory CD8$^+$ T cells from 11.5 to 42.4% ($P = 0.006$).

Simultaneously we observed changes in the T cell subsets representing a activated/differentiated phenotype (HLA-DR$^+$, CD27$^-$, CD28$^-$, and CD57$^+$); they changed significantly and showed a strong age-dependent increase in both CD4$^+$ and CD8$^+$ T cells. The subset changing the most with age was the CD27$^-$ effector T cells, which increased from 4.4% to 7.5% ($P = 0.005$) for CD4$^+$ and from 8.9% to 39.7% ($P < 0.001$) for CD8$^+$ T cells (Table 3).

The age-dependent increase of differentiated/senescent/activated T cells was most apparent when analyzing their proportional changes, but several cell subsets (CD27$^-$, CD57$^+$, and naïve) also increased significantly in absolute numbers (Supporting Information Table S2).

When fitting a weighted regression line (which is robust against the heavily skewed distribution of data), it was found that the linear models reflect the continuous changes over age observed in differentiated/senescent/activated T cell subsets. Interestingly, the individuals in this cohort seem to segregate into a group with very high proportions of differentiated/activated (CD27$^-$, CD57$^+$) CD4$^+$ T cells, which further increase with age, while the majority of individuals maintain low frequencies with only minor or no age-dependent changes in these subsets (Fig. 2).

The median proportion of total Tregs was 7.9% of CD4$^+$ T cells, which was not affected significantly by age (Table 3). However, when analyzing the distribution of naïve and memory Tregs based on CD45RA expression, we observed a strong increase in memory Tregs from 68.3% to 89.8% of Tregs ($P < 0.001$) with age, while the naïve CD45RA$^+$ Tregs decreased correspondingly from 31.7% to 10.2% ($P < 0.001$).

The total number of gamma-delta TCR ($\gamma\delta$-TCR) T cells remained unchanged with a median of 3.7% of all T cells (Table 3). The very rare and largely unknown $\gamma\delta$-TCR CD4$^+$ subset decreased from 0.5% to 0.2% within the $\gamma\delta$-TCR T cell population ($P = 0.012$). The majority of $\gamma\delta$-TCR was CD4$^-$CD8$^-$ amounting to no <84.7%, followed by 14.6% $\gamma\delta$-TCR CD8$^+$ T cells.

The median proportion of B cells within lymphocytes was 11.3%, and showed no significant change with age (Table 3). Similarly, neither naïve nor memory B cells changed significantly with age and reached medians of 54.6% and 21%, respectively. We observed a marked decrease in MZ B cells from a median of 12.3% to 7.6%, although this difference did not reach statistical significance (rho = −0.32)

Second Cohort Validation of Age-Dependent Differences

To test whether our findings from the first cohort can be easily validated, we recruited another 46 healthy individuals that matched the first cohort with regard to age and gender distribution (Table 1). Importantly, a different operator performed sample staining and measurements, and thus the second cohort served to assess the robustness of our sample handling and staining procedures and to assess the reproducibility of reference values for all subsets as determined by the standardized "ONE Study" flow cytometric profiling. The results of the validation are shown in Table 4 (and Figs. 3 and 4), and they could, with very few exceptions, reproduce and validate the results of the first cohort. The mean values of only 3 subsets (Tregs, $\gamma\delta$-TCR CD4$^+$ T cells and non-switched memory B cells) could not be considered equivalent between

Table 4. Comparison of medians and interquartile ranges between the two cohorts

VALIDATION, BOTH COHORTS	MEDIAN PROPORTION (INTERQUARTILE RANGE)				MEDIAN ABSOLUTE COUNT (INTERQUARTILE RANGE)			
CELL SUBSET	FIRST COHORT	SECOND COHORT	RTOST (SD)	RLM (AGE)	FIRST COHORT	SECOND COHORT	RTOST (SD)	RLM (AGE)
$CD45^+$ Leucocytes					5610 (4950–6665)	5980 (5290–6675)	Eq. (1)	not diff.
Granulocytes	58.5 (50.7–63.1)	58.1 (55.3–62.7)	Eq. (1)	Not diff.	3094 (2528–4044)	3462 (2995–4200)	Eq. (1)	**0.033**
$CD14^+$ Monocytes	7.1 (5.7–8.1)	7.2 (6–8.7)	Eq. (1)	Not diff.	426 (306–512)	425 (346–548)	Eq. (1)	Not diff.
$CD14^{high}$ $CD16^-$	85.7 (82.7–89.7)	88.9 (**86.2***−91.2)	Eq. (**1.5**)	Not diff.	382 (256–445)	361 (303–483)	Eq. (1)	Not diff.
$CD14^{high}$ $CD16^+$	5.8 (4.3–8.3)	5 (3.7–6.3)	Eq. (**1.5**)	Not diff.	24 (17–32)	21 (14–33)	Eq. (1)	Not diff.
$CD16^{high}$	6 (4.2–7.9)	4.5* (3.2* −5.7*)	Eq. (**1.5**)	**0.035**	23 (15–34)	19 (14–28)	Eq. (1)	Not diff.
Dendritic Cells	0.8 (0.6–1)	0.7 (0.5–0.9)	Eq. (1)	Not diff.	42 (36–63)	42 (30–54)	Eq. (1)	Not diff.
pDCs	0.1 (0.1–0.2)	0.1 (0.1–0.2)	Eq. (**1.5**)	Not diff.	8 (4–12)	10 (6–14)	Eq. (1)	Not diff.
mDCs	0.7 (0.4–0.9)	0.5 (0.4–0.7)	Eq. (**1.5**)	Not diff.	37 (26–52)	32 (23–40)	Eq. (1)	Not diff.
$CD16^+$ mDCs	75.9 (62.1–82)	71.4 (63.3–79.2)	Eq. (1)	Not diff.	30 (16–40)	23 (14–31)	Eq. (1)	Not diff.
MDC1	23.2 (17.1–36.6)	27.5 (20.3–35.5)	Eq. (1)	Not diff.	9 (6–11)	8 (7–11)	Eq. (1)	Not diff.
MDC2	0.9 (0.5–1.4)	1 (0.7–1.4)	Eq. (1)	Not diff.	0 (0–1)	0 (0–0)	Eq. (1)	Not diff.
Lymphocytes	32.6 (27.6–40.9)	33.8 (28.6–36.4)	Eq. (1)	Not diff.	1875 (1651–2362)	2000 (1544–2249)	Eq. (1)	Not diff.
$CD56^+$ NK cells	13.2 (8.6–15.9)	10.4 (8.1–14.9)	Eq. (**1.5**)	Not diff.	230 (188–300)	207 (146–310)	Eq. (1)	Not diff.
$CD56^{dim}$	94.9 (93.2–96.8)	95.2 (91.5–97.4)	Eq. (1)	Not diff.	213 (172–289)	191 (136–303)	Eq. (1)	Not diff.
$CD56^{high}$	5.1 (3.2–6.8)	4.8 (2.6–8.5)	Eq. (1)	Not diff.	11 (8–14)	10 (7–16)	Eq. (1)	Not diff.
$CD3^+$ T cells	69.5 (66.3–76.3)	73.8 (67.6–76.7)	Eq. (**1.5**)	Not diff.	1342 (1055–1697)	1399 (1053–1678)	Eq. (1)	Not diff.
$CD4^+$	61 (54.9–67.1)	60.7 (54.9–69.6)	Eq. (1)	Not diff.	799 (637–1050)	885 (615–1049)	Eq. (1)	Not diff.
$HLA–DR^+$	29.9 (17.4–45.3)	7.1 (5.9–11.1)	Eq. (1)	Not diff.	120 (57–209)	53 (44–81)	Eq. (1)	Not diff.
$CD45RA^+$	C0.3 (0.2–0.7)	0.2 (0.2–0.3)	Eq. (1)	Not diff.	2 (1–6)	2 (1–3)	Eq. (1)	Not diff.
$CD45RA^-$	7.5 (5.2–12.4)	7 (5.6–9)	Eq. (1)	Not diff.	61 (45–97)	51 (43–80)	Eq. (1)	Not diff.
Naïve	36.1 (23.5–45.7)	37.6 (27.4–52.3)	Eq. (1)	Not diff.	289 (153–472)	293 (212–467)	Eq. (1)	Not diff.
CM	39.5 (33.2–47.4)	44.1 (35.4–51.7)	Eq. (1)	Not diff.	317 (254–442)	334 (259–459)	Eq. (1)	Not diff.
EM	13.9 (11.2–18.7)	14.3 (10.9–19.8)	Eq. (1)	Not diff.	119 (86–161)	124 (83–157)	Eq. (1)	Not diff.
TEMRA	0.4 (0.2–1.4)	0.2 (0.2–0.5)	Eq. (1)	Not diff.	4 (1–12)	**2*** (**1–4**)	Eq. (1)	**0.020**
$CD27^-$	6.5 (4.5–12.7)	6.6 (4.1–10.8)	Eq. (1)	Not diff.	52 (37–86)	51 (31–90)	Eq. (1)	Not diff.
$CD28^-$	0.8 (0.1–3.6)	0.4 (0.1–2.8)	Eq. (1)	Not diff.	6 (1–24)	4 (1–30)	Eq. (1)	Not diff.
$CD57^+$	1.9 (1–6.7)	1.4 (**0.5*** −4.1)	Eq. (1)	Not diff.	15 (9–42)	10 (**4*** −39)	Eq. (1)	Not diff.
Tregs	**7.9 (7.2–9)**	**5.2* (3.6* –7.6*)**	**not eq.**	**<0.001****	66 (47–96)	39 (26–63)	Eq. (1.5)	Not diff.
$CD8^+$	32.5 (27.6–38.5)	33.4 (25.8–36.8)	Eq. (1)	Not diff.	438 (334–527)	419 (315–542)	Eq. (1)	Not diff.
$HLA–DR^+$	8 (5.6–13.5)	20.8 (13.7–37.7)	Eq. (**1.5**)	Not diff.	64 (48–104)	78 (46–158)	Eq. (1)	Not diff.
$CD45RA^+$	11.8 (3.5–19.5)	6.7 (3.8–12.6)	Eq. (1)	Not diff.	38 (14–82)	23 (16–51)	Eq. (1)	Not diff.
$CD45RA^-$	16.4 (10.5–23.6)	12.9 (8.4–23.4)	Eq. (1)	Not diff.	71 (38–114)	54 (27–102)	Eq. (1.5)	Not diff.
Naïve	26.2 (13.3–42.5)	33.7 (20.8–45.9)	Eq. (1)	Not diff.	111 (44–205)	127 (73–190)	Eq. (1)	Not diff.
CM	11.7 (9–16.8)	10.7 (8.1–14.7)	Eq. (1)	Not diff.	48 (38–75)	43 (30–66)	Eq. (1)	Not diff.
EM	34.1 (25.6–42.8)	30 (22.9–38.6)	Eq. (1)	Not diff.	155 (84–226)	120 (87–176)	Eq. (1)	Not diff.

TABLE 4. *Continued*

VALIDATION, BOTH COHORTS	MEDIAN PROPORTION (INTERQUARTILE RANGE)				MEDIAN ABSOLUTE COUNT (INTERQUARTILE RANGE)			
CELL SUBSET	FIRST COHORT	SECOND COHORT	RTOST (SD)	RLM (AGE)	FIRST COHORT	SECOND COHORT	RTOST (SD)	RLM (AGE)
TEMRA	18.6 (8.2–41.8)	20.1 (9.8–28.9)	Eq. (1)	Not diff.	62 (36–154)	70 (35–137)	Eq. (1)	Not diff.
CD27^{-}	21.6 (7.3–38.9)	21 (6.6–34.4)	Eq. (1)	Not diff.	78 (28–162)	83 (27–155)	Eq. (1)	Not diff.
CD28^{-}	27.9 (14.9–46.7)	30.6 (12.8–46.7)	Eq. (1)	Not diff.	102 (57–201)	127 (58–183)	Eq. (1)	Not diff.
CD57^{+}	24.6 (12.6–42)	24.2 (8.7–39.9)	Eq. (1)	Not diff.	98 (51–187)	89 (36–154)	Eq. (1)	Not diff.
TCR–gd	3.8 (2.5–5.9)	4 (2.6–6.2)	Eq. (1)	Not diff.	49 (30–84)	60 (33–88)	Eq. (1)	Not diff.
CD4^{+}	0.4 (0.2–0.9)	0.7 (**0.3*** −1.1)	**not eq.**	Not diff.	0.2 (0.1–0.4)	0.3 (0.2–0.5)	**not eq.**	Not diff.
CD8^{+}	14.6 (10.4–24.8)	18 (11.8–23.9)	Eq. (1)	Not diff.	8 (4–16)	11 (4–17)	Eq. (1)	Not diff.
CD4^{-} CD8^{-}	84.7 (73.9–88.5)	80.6 (73.1–87.9)	Eq. (1)	Not diff.	38 (22–69)	46 (29–70)	Eq. (1)	Not diff.
CD4^{+}:CD8^{+}	–	–			1.9 (1.4–2.4)	1.8 (1.5–2.7)	Eq. (1)	Not diff.
CD19^{+} B cells	11.3 (9–13.5)	10.6 (7.4–12.8)	Eq. (**1.5**)	Not diff.	205 (149–289)	203 (122–270)	Eq. (1)	Not diff.
Naïve B cells	53.6 (39.7–64.7)	55.1 (47.3–65.5)	Eq. (1)	Not diff.	43 (24–69)	106 (71–162)	Eq. (1)	Not diff.
Transitional B cells	2.1 (1.3–3.4)	2.8 (1.6–4.2)	Eq. (**1.5**)	Not diff.	2 (1–4)	6 (2–10)	Eq. (1.5)	Not diff.
MZ B cells	12.5 (7.8–15.6)	14.3 (8.2–21.2)	Eq. (**1.5**)	Not diff.	8 (5–23)	23 (14–40)	Eq. (1)	Not diff.
Memory B cells	21 (16.8–29.7)	18.8 (14.6–24.2)	Eq. (1)	Not diff.	18 (11–30)	**31*** (**22***–**49***)	Eq. (1.5)	Not diff.
Class-switched B cells	15.9 (10–20.6)	14.2 (10.8–19)	Eq. (1)	Not diff.	14 (8–21)	24 (17–38)	Eq. (1)	Not diff.
Nonswitched B cells	**6** (**4.4–9.1**)	**3.9*** (**2.3*** –**5.6***)	**Not eq.**	**0.003**	**5** (**3–10**)	**8*** (**4***–**10***)	**not eq.**	**0.005**
Plasmablasts	1.2 (0.8–2.2)	1.1 (0.5–2)	Eq. (1)	Not diff.	1 (1–2)	2 (1–3)	Eq. (1.5)	Not diff.
CD21^{low} B cells	6.2 (4.2–10.2)	8.1 (5.6–12.8)	Eq. (**1.5**)	Not diff.	5 (3–9)	15 (10–23)	Eq. (1)	Not diff.

Equivalence of cell population means was tested by robust two one-sided tests (rTOST). Subset means that differed <1 standard deviation (calculated from the first cohort) between the two cohorts were considered equivalent. Potential equivalence is established at a critical mean difference of 1.5 SD_{1st} units.

*Significantly different quartiles are highlighted. Age as covariate was tested robust linear models (rLM) for each subset. *P*-values were not adjusted for multiple comparisons.

**In the second cohort the Treg subset was analysed with a different staining panel. The subset denominator for proportional results is listed in Table 2.

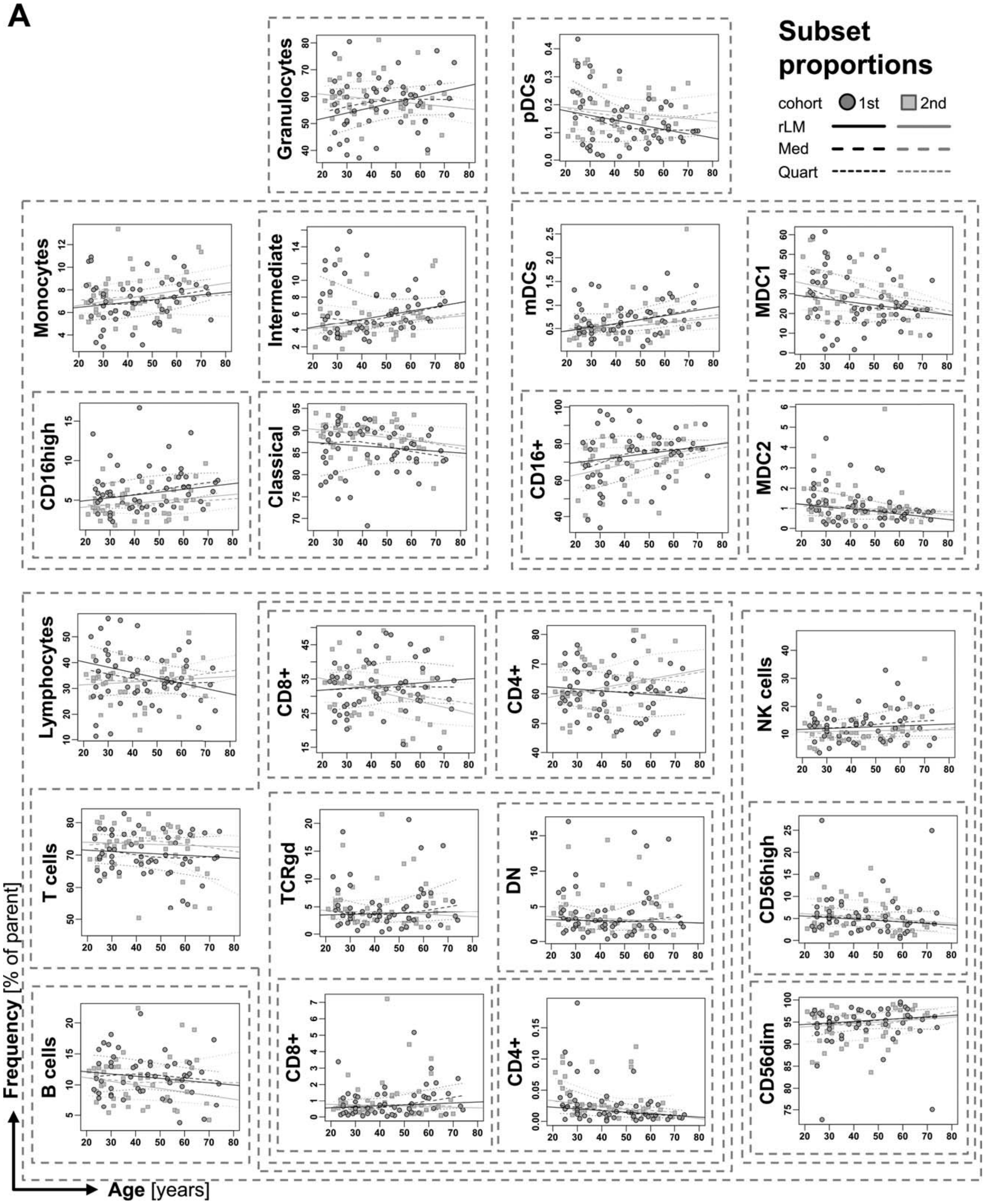

Figure 3. Age dependency of major leucocyte subsets. The cell subset dependency with age is illustrated by smoothed curves for medians (Med) and interquartile ranges (Quart), as well as by robust linear regression lines (rLM) in proportional (**A**) and absolute data (**B**). Grouped subsets are encircled by dashed boxes. For the individual subset definitions see Table 2. *P*-values for proportions and absolute numbers of cell populations, which significantly correlate with age, are listed in Table 3 and Supporting Information Table S2, respectively.

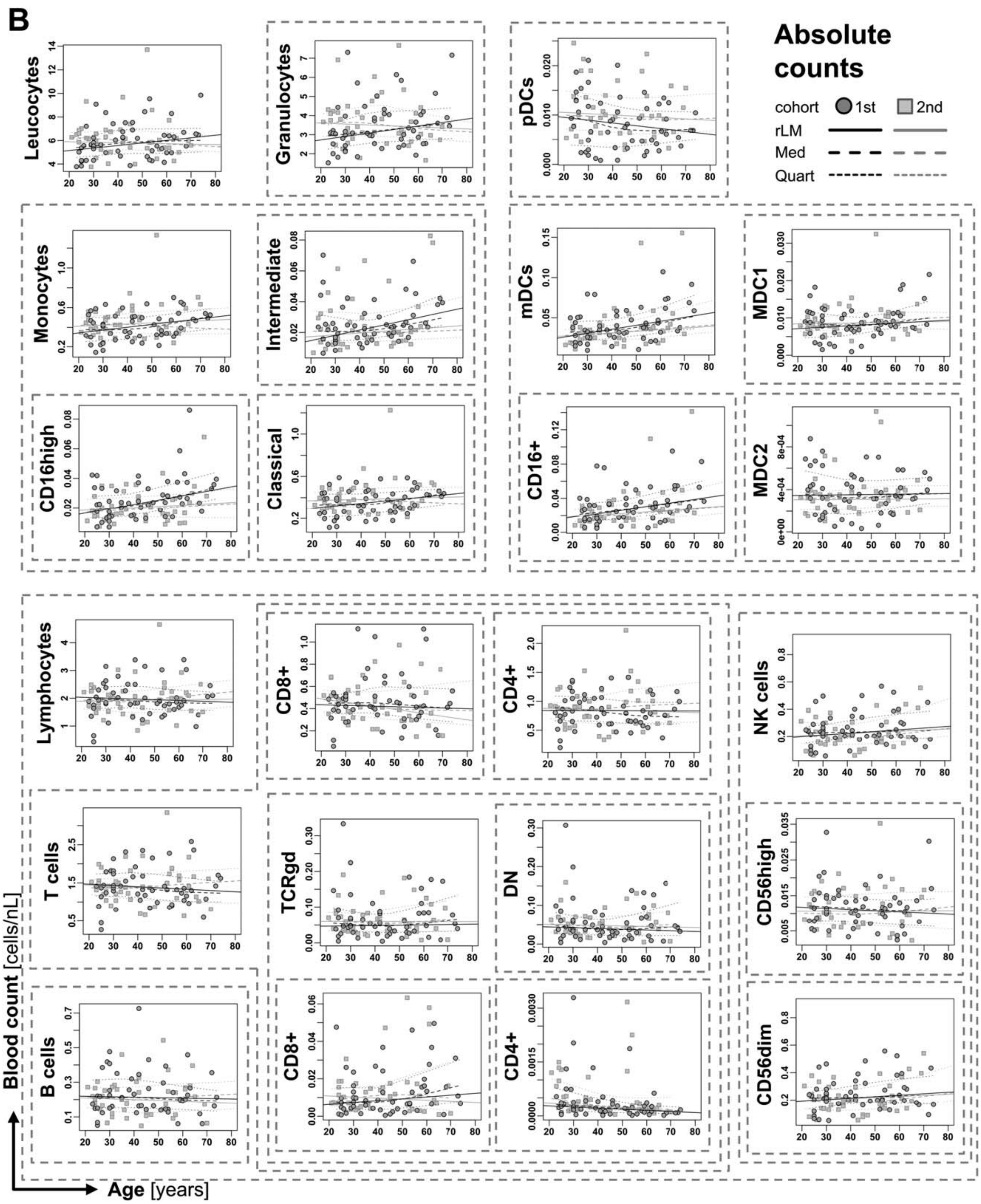

Figure 3. (Continued).

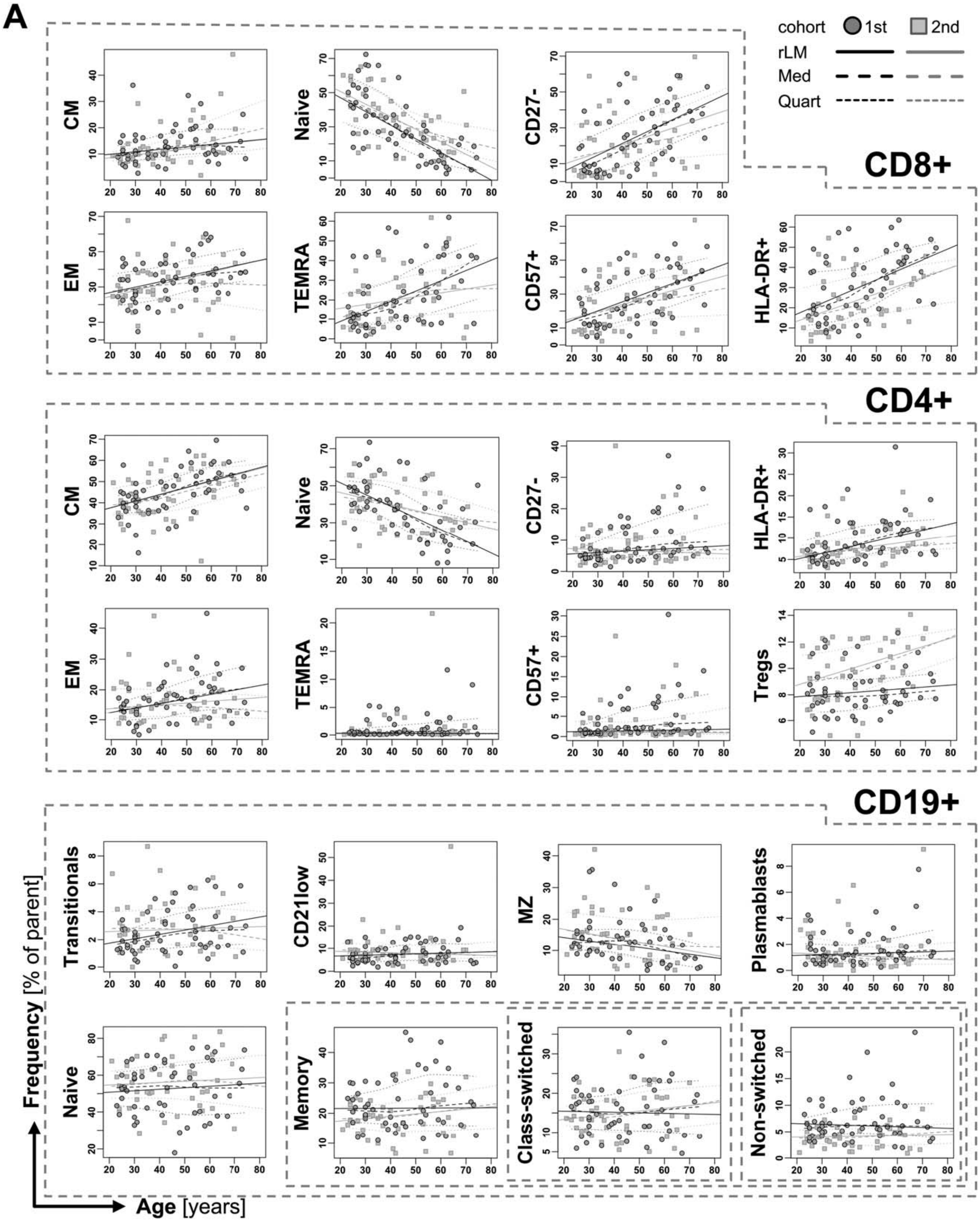

Figure 4. Age dependency of B and T cell subsets. The age dependency is illustrated by smoothed curves for medians (Med) and interquartile ranges (Quart), as well as by robust linear regression lines (rLM) in proportional (**A**) and absolute data (**B**). The T cell subsets shown (except Tregs and Naive) represent differentiated and activated subsets. Grouped subsets are encircled by dashed boxes. For the individual subset definitions see Table 2. *P*-values for proportions and absolute numbers of cell populations, which significantly correlate with age, are listed in Table 3 and Supporting Information Table S2, respectively.

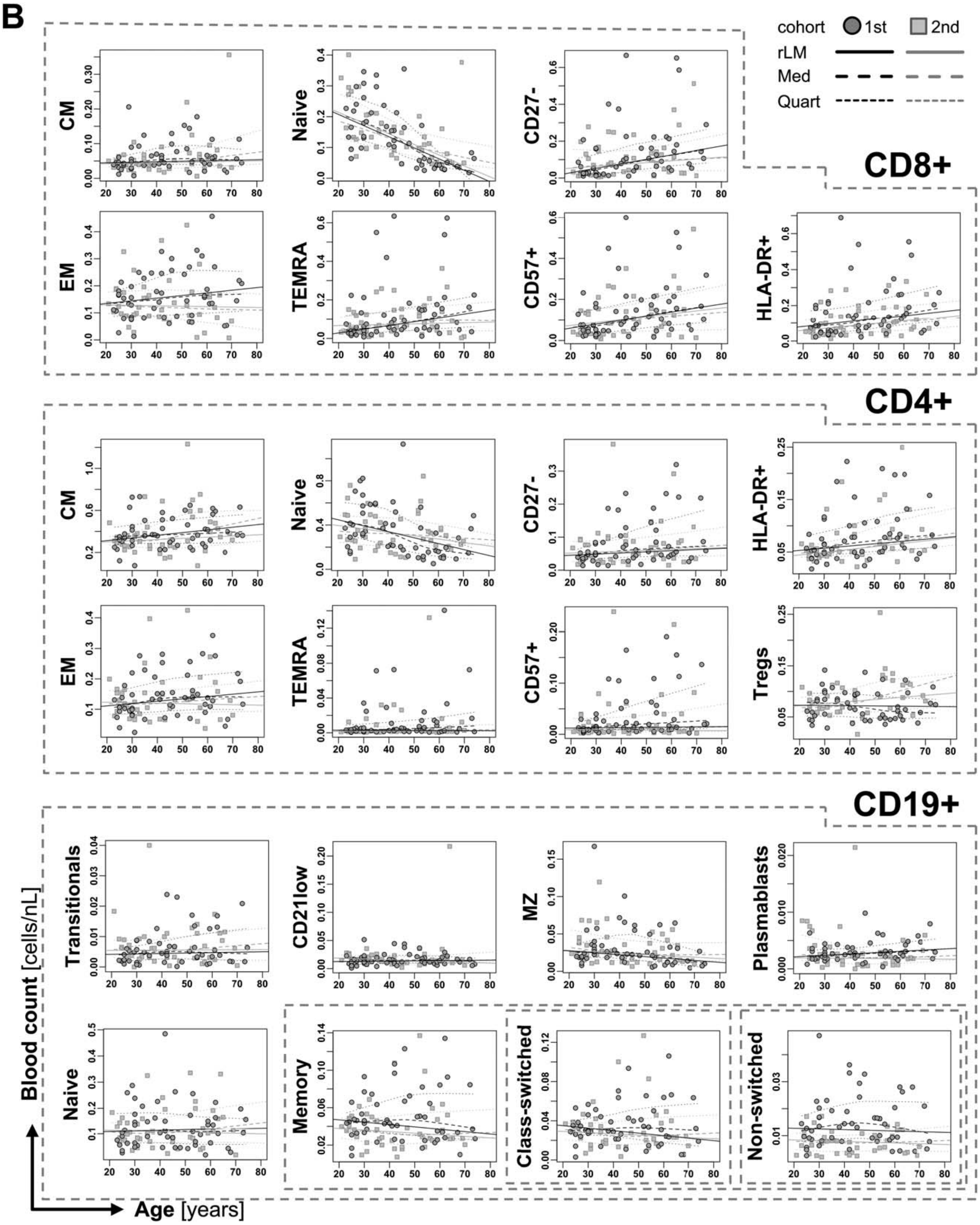

Figure 4. (Continued).

the first and second cohort. However, the non-equivalence of Treg counts are likely attributed to the different staining protocols used for the two cohorts (see methods section). In addition, 5 subsets (granulocytes, $CD16^{high}$ monocytes, $CD4^+$ T_{EMRA} T cells, Tregs, and nonswitched memory B cells) showed a difference in their age-dependent regression lines.

A significant correlation with age was confirmed for most of the tested subsets that had previously been identified in the first cohort. Correlation did not reach statistical significance in the second cohort for MDC2, T_{EM} ($CD4^+$ and $CD8^+$), $CD4^+$ and $CD4^+$ $CD27^-$, $CD4^+$ $CD57^+$ T cells. Interestingly, the MZ B cell correlation, which was noted as an age-dependent decrease in the first cohort, lost significance through adjustment for multiple comparisons, but showed a significant association with age in the second validation cohort.

Gender Influence on Leucocyte Composition

To achieve sufficient power we used combined data from both the first and second cohort to test whether we can also reveal gender influence on leucocyte subset composition and age-dependent changes. The results are shown in Table 5 and Figure 5.

Innate immune cell subsets showed no significant gender-specific distributions. The only difference observed was a gender-specific age dependency measured by a significant difference in robust regression slopes. Males showed a stronger tendency toward monocyte ($P < 0.01$) and $CD16^+$ mDC ($P < 0.001$) accumulation, with loss of MDC1 ($P < 0.001$) and MDC2 ($P < 0.001$), while females had fewer pDCs ($P < 0.001$) and $CD14^{high}$ $CD16^-$ monocytes ($P < 0.001$) with age.

We observed significant differences within the lymphocyte subsets. A significantly larger proportion of $CD8^+$ T cells (34.5%) was observed in samples collected from male healthy individuals, versus females (29.5%; $P = 0.031$), but this difference was not significant when considering absolute cell numbers. However, the proportion of $CD4^+$ T cells was slightly higher in females (64.6% of all T cells), versus males (58.3%; $P = 0.026$; Fig. 5). This was also apparent when calculating the $CD4^+$:$CD8^+$ ratios which were 2.1 in females and 1.7 in males ($P = 0.026$). In addition, the proportion of differentiated/activated HLA-DR^+, $CD27^-$, $CD28^-$, and especially $CD57^+$ cells within the $CD8^+$ T cell compartment were significantly higher in samples from male volunteers ($P = 0.026$; $P = 0.042$; $P = 0.034$; $P = 0.026$; Fig. 5). The same pattern was seen in the $CD4^+$ compartment, but only the $CD4^+$ HLA-DR^+ T cells achieved statistical significance ($P = 0.036$).

Considering differences in age-dependent changes, $CD8^+$ T cells gain HLA-DR expression at significantly different rates in females and males ($P < 0.001$). As shown in Figure 5, young males already show high frequencies of HLA-DR^+ $CD8^+$ T cells which further increase slightly with advancing age, whereas females have significantly fewer activated $CD8^+$ T cells at young age ($P < 0.0001$) and the HLA-DR^+ $CD8^+$ proportion increases during aging and reach levels similar to males.

We also observed some variation within $\gamma\delta$-TCR T cell subsets. The $\gamma\delta$ TCR $CD4^+$ T cells were more prevalent in females at 0.8% of all $\gamma\delta$ TCR cells, versus 0.3% in males, but this difference was not significant after P values adjustment. Females showed a steep age-dependent decrease in $\gamma\delta$ TCR $CD4^+$ cells that was not present in males ($P < 0.001$; Fig. 5); this could be attributed to differences solely in the first age decade ($P = 0.008$).

The B cell compartment showed no significant gender-specific distribution after adjusting for multiple comparisons. Notably, transitional B cells tend to be higher in females than in males with an adjusted P values of 0.059 (non-adjusted 0.015). Age-dependency of $CD21^{low}$ B cells showed significantly different and inverse relationships for males and females ($P < 0.001$). $CD21^{low}$ B cell proportions increased during aging in males, whereas they decreased in females (Fig. 5).

Discussion

Age affects the immune system to a degree that has led to concepts such as immunosenescence and inflammaging. Understanding and utilizing these age-effects might improve management of infections, autoimmune disease, immunotherapy and even prevention of cancer. Multiparametric flow cytometry represents an ideal method to reveal age and gender effects on immune cell composition, but its low reproducibility has hampered its use in the past. In contrast to previous studies, which often only studied changes of particular immune cell subsets, we used a recently established standardized set of 6 flow panels capturing over 50 different leucocyte subsets to verify previously described and identify novel age- and gender-dependent alterations of innate and adaptive immune cell subpopulations simultaneously. Furthermore, the identified patterns of altered cell subsets were largely validated in a second cohort of healthy individuals although the multi-parametric approach required corrections for multiple comparisons.

Only a few cell subsets of the innate immune system showed significant alterations with age. An increase (spearman correlation coefficient >3.0) was observed in absolute numbers of monocytes – especially for the $CD14^+$ $CD16^{high}$ subset, granulocytes and DCs. Several groups have described an age-dependent increase of $CD14^+$ $CD16^{high}$ (non-classical) monocytes (19,43,44). Nyugen et al. have shown that monocyte cytokine-production in the elderly is only decreased in response to stimulation, suggesting an unspecific up-regulation combined with failing responsiveness. Our data does not confirm an up-regulation of $CD14^+$ $CD16^{high}$ monocytes as reported by other studies. It is possible that the individual variation and heteroscedasticity within this subset makes it more suitable for statistical testing between a young and elderly group as performed in those studies.

The clinical consequences of aging DCs on the senescent immune system are largely unknown. Studies have shown altered cytokine producing profiles consistent with inflammaging (45). In our study the numbers of dendritic cells are largely unchanged with age, supporting the general theory of an age-conserved innate immune system.

Table 5. Medians and interquartile range of the males and females in combined cohorts and with both absolute (cells/μL) and proportional cell numbers

GENDER, BOTH COHORTS	MEDIAN PERCENTAGE (INTERQUARTILE RANGE)			MEDIAN ABSOLUTE COUNT (INTERQUARTILE RANGE)		
CELL SUBSET	MALE	FEMALE	P_{CWMW} (P_{RLM})	MALE	FEMALE	P_{CWMW} (P_{RLM})
CD45$^+$ Leucocytes	–	–		5965 (4995–6450)	5840 (5180–6885)	
Granulocytes	58.9 (54.6–64.7)	57.6 (51.6–62.2)		3208 (2811–4130)	3355 (2602–4158)	
CD14$^+$ Monocytes	7.4 (5.9–9.1)	6.9 (5.9–8)	ns (**)	447 (338–566)	385 (318–502)	ns (**)
CD14^{high} CD16$^-$	86.9 (83.5–90.2)	88.9 (84.3–90.9)	ns (***)	387 (276–493)	351 (274–448)	ns (**)
CD14^{high} CD16$^+$	5.5 (4.2–7.7)	4.9 (4–7.6)		24 (16–40)	20 (14–26)	ns (***)
CD16^{high}	5.7 (4.3–7.6)	4.8 (3.4–6.2)		25 (17–35)	20 (13–26)	
Dendritic Cells	0.7 (0.5–0.9)	0.8 (0.5–1)		41 (32–59)	43 (36–61)	
pDCs	0.1 (0.1–0.2)	0.1 (0.1–0.2)	ns (***)	8 (5–13)	9 (6–13)	ns (*)
mDCs	0.6 (0.4–0.8)	0.6 (0.4–0.9)		34 (23–47)	35 (25–48)	
CD16$^+$ mDCs	72.2 (59.9–80.4)	76.4 (64.1–80.2)	ns (***)	25 (14–36)	26 (17–37)	
MDC1	26.4 (18.8–38.4)	23 (18.7–34.7)	ns (***)	8 (6–12)	8 (7–10)	
MDC2	1 (0.6–1.9)	0.9 (0.5–1.3)	ns (***)	0 (0–1)	0 (0–0)	
Lymphocytes	32.4 (27.1–36.7)	33.9 (29–41)		1832 (1543–2212)	1957 (1695–2381)	ns (**)
CD56$^+$ NK cells	12.9 (9.3–16.2)	10.5 (8–15.8)		235 (186–311)	201 (151–298)	
CD56^{dim}	95.9 (93.8–97.6)	94.5 (91.6–96.7)		229 (172–303)	195 (137–286)	
CD56^{high}	4.1 (2.4–6.2)	5.5 (3.3–8.5)		10 (7–14)	12 (8–17)	
CD3$^+$ T cells	69.8 (66.4–76.8)	73.5 (67.8–76.3)	ns (*)	1284 (1037–1565)	1417 (1208–1780)	ns (**)
CD4$^+$	**58.3 (52.2–62.4)**	**64.6 (58.7–70.6)**	**0.026** (ns)	746 (580–969)	901 (688–1169)	
HLA–DR$^+$	**15.4 (8.8–40.7)**	**10.1 (6.5–23.8)**	**0.036** (ns)	92 (53–154)	66 (45–110)	
CD45$^+$	0.2 (0.2–0.7)	0.2 (0.2–0.3)		2 (1–4)	2 (1–4)	
CD45$^-$	**8.3 (6.1–12.1)**	**6.7 (4.6–8.4)**	**0.041** (ns)	60 (46–97)	52 (42–77)	
Naive	35.6 (24.1–43.6)	38.8 (28.7–49.5)		260 (158–373)	356 (209–499)	ns (***)
CM	43.1 (34.6–53.3)	40.4 (32.7–47.4)		308 (232–432)	334 (278–462)	
EM	14.5 (11.1–19.7)	13.7 (10.5–19.1)		111 (72–155)	131 (99–165)	
TEMRA	0.3 (0.2–1.2)	0.3 (0.2–0.7)		2 (1–8)	2 (1–7)	
CD27$^-$	6.6 (4.6–13.5)	6 (4–10.8)	ns (*)	58 (34–94)	48 (35–87)	
CD28$^-$	0.7 (0.2–3.6)	0.5 (0.1–2.8)		5 (1–24)	4 (1–32)	
CD57$^+$	2.1 (0.9–6.6)	1.3 (0.8–3.1)		17 (5–42)	12 (6–37)	
Tregs	7.3 (5.2–8.5)	7.5 (5–9)		48 (36–79)	62 (37–83)	
CD8$^+$	**34.5 (30.5–38.5)**	**29.5 (24.8–34.6)**	**0.031** (**)	445 (332–521)	419 (299–542)	ns (**)
HLA–DR$^+$	**13.9 (8.3-32.2)**	**11 (6.3–17)**	**0.026** (***)	89 (53–145)	59 (39–83)	
CD45$^+$	**12 (5.2–20.8)**	**6 (3.3–12.7)**	**0.040** (ns)	45 (22–82)	22 (11–41)	
CD45$^-$	**18.4 (11.7–24.9)**	**10.9 (8.3–19.3)**	**0.040** (ns)	79 (46–132)	44 (27–86)	
Naïve	25.3 (13.3–41.2)	33.3 (22.1–45.7)		106 (55–176)	133 (60–216)	ns (***)
CM	10.5 (8.1–16)	12 (8.6–17.5)	ns (***)	49 (34–69)	46 (33–64)	
EM	34.1 (23.8–44)	31.7 (23.7–38.4)	ns (**)	149 (86–239)	133 (86–181)	
TEMRA	21.9 (11–38.5)	16.8 (7.6–29.5)	ns (***)	84 (38–166)	59 (35–134)	
CD27$^-$	**24.3 (10.1–41.6)**	**16.9 (5.7–30.7)**	**0.042** (ns)	111 (33–198)	60 (21–122)	
CD28$^-$	**36.1 (20.4–48.8)**	**26.3 (12.8–38.1)**	**0.034** (ns)	164 (66–242)	95 (52–165)	
CD57$^+$	**30.8 (16.6–44.3)**	**19.5 (9.5–30.5)**	**0.026** (ns)	140 (53–215)	83 (36–143)	

TABLE 5. *Continued*

GENDER, BOTH COHORTS	MEDIAN PERCENTAGE (INTERQUARTILE RANGE)			MEDIAN ABSOLUTE COUNT (INTERQUARTILE RANGE)		
CELL SUBSET	MALE	FEMALE	P_{CWMW} (P_{RLM})	MALE	FEMALE	P_{CWMW} (P_{RLM})
TCRgd	4.5 (2.9–7.5)	3.3 (2–5.7)		64 (40–88)	47 (29–84)	ns (*)
$CD4^+$	0.4 (0.2–0.7)	0.8 (0.3–1.6)	ns (***)	0.2 (0.1–0.3)	0.3 (0.1–0.8)	ns (***)
$CD8^+$	15.9 (10.9–20.8)	17.2 (10.5–28.9)		10 (5–16)	8 (4–18)	ns (***)
$CD4^-$ $CD8^-$	83.6 (78.7–88.4)	81.2 (69.7–87.8)	ns (**)	49 (32–75)	38 (21–67)	ns (***)
$CD4^+$:$CD8^+$	–	–		**1.7 (1.4–2)**	**2.1 (1.7–2.8)**	**0.026** (ns)
$CD19^+$ B cells	11.1 (7.8–13.3)	10.9 (8.8–13.1)	ns (**)	206 (141–271)	203 (145–275)	
Naïve B cells	53.9 (41–65.9)	54.2 (43.3–65)		58 (33–144)	70 (45–108)	ns (**)
Transitional B cells	2.1 (1.3–3)	3 (1.6–4.4)		2 (1–6)	3 (2–6)	
MZ B cells	12.8 (8.3–16.7)	12.5 (7.8–20.8)		13 (7–30)	16 (7–34)	
Memory B cells	19.3 (15.1–30)	20.8 (16–24.5)		25 (15–37)	27 (16–46)	
Class–switched B cells	14.5 (10.6–23)	14.9 (10.7–19)	ns (*)	18 (10–28)	20 (12–31)	
Non–switched B cells	4.9 (3.1–7.2)	5.1 (3.7–6.3)		5 (3–8)	6 (4–12)	
Plasmablasts	1.2 (0.8–2)	1.1 (0.5–2.2)	ns (**)	1 (1–2)	1 (1–3)	
CD21low B cells	7.5 (4.8–13.3)	7 (4.7–10)	ns (***)	9 (5–16)	11 (4–17)	ns (***)

Gender difference was tested by Cliffs analogue of the nonparametric Wilcoxon-Mann-Whitney statistic (cWMW) on age-adjusted residuals. When an association with age, that is, slopes of robust regression lines (rLM), significantly differed between males and females, age was not taken in to account as a covariate. Only *P*-values < 0.05/levels of significance (*$P < 0.05$. ** $P < 0.01$. *** $P < 0.001$) after multiple comparison correction are reported. The subset denominator for proportional results is listed in Table 2.

The absolute number of mDCs and especially the pro-inflammatory $CD16^+$ mDCs showed a tendency (not statistically significant) toward an age-dependent increase. The $BDCA3^+$ MDC2 subset declined with age and was the only dendritic cell subset with a significant correlation to age ($P = 0.033$). Age related change to mDCs is still unclear, since they have been reported to either decline or remain unchanged (44,46–49). A study also showed increasing MDC2 numbers, opposite to our findings (44). Different methods and data analyses may explain some of this variation, that is, Orsini et al. (47) found a decrease in mDC numbers, but made a separate analysis for >20 year-old participants where there was no significant decline. A consensus on the identification and gating strategy of mDCs is another unresolved matter. Here, we defined mDCs as $lin^-HLA\text{-}DR^+$ $CD11c^+$, and to our knowledge, only one other group used the same surface markers to test for age dependency; this group also showed no significant change to mDCs, in agreement with our results (49). From our data we found that the plasmacytoid DCs (pDCs) did not change significantly with age. Other studies report a decline in pDCs, especially in absolute numbers (44,46–48,50). Again the absence of an international consensus on surface markers used to identify the subset, standardized staining sample handling, staining procedures, and gating strategy may explain the observed differences. In addition, we noted a tendency toward declining proportions and numbers. Analysing a low frequency subset like pDCs in a larger cohort might have revealed a statistically significant reduction during aging.

The number of NK cells and their $CD56^{dim}$ and $CD56^{high}$ subsets also was conserved during aging. Other studies have reported similar findings but with shift from regulatory $CD56^{dim}$ to the cytotoxic $CD56^{high}$ subset that we did not observe (44,51,52). Even the $CD56^{dim}$:$CD56^{bright}$ ratio as shown by some studies was not significantly altered with older age (data not shown).

Our data show that the T cell compartment undergoes substantial change during the 40–50 year life span examined here. The T cell dominance of age-correlated changes is also visible in both heatmaps, where most of the subsets contributing to the age separation belonged to the T cell compartment (Figures 1A and 1B). With advancing age, the sizes of the naïve $CD4^+$ and $CD8^+$ T cell subsets are heavily reduced, and the $CD8^+$ compartment is instead dominated by effector memory (T_{EM}) and effector memory $CD45RA^+$ (T_{EMRA}) T cells, and the $CD4^+$ compartment by central memory (T_{CM}) T cells (22,53,54). These variations of subset dominance could be attributed to the different functionalities of $CD4^+$ and $CD8^+$ T cells, but could also result from different aging processes. Gender seemed to affect the overall $CD4^+$ and $CD8^+$ distribution with females having more $CD4^+$ T cells and males more $CD8^+$ T cells, resulting in a markedly higher $CD4^+$:$CD8^+$ ratio in females relative to previous reports (4,55,56).

T cells are presumed to follow a certain surface-marker differentiation, first losing CD45RA expression, then CCR7 (or CD62L), and finally regaining CD45RA (57).

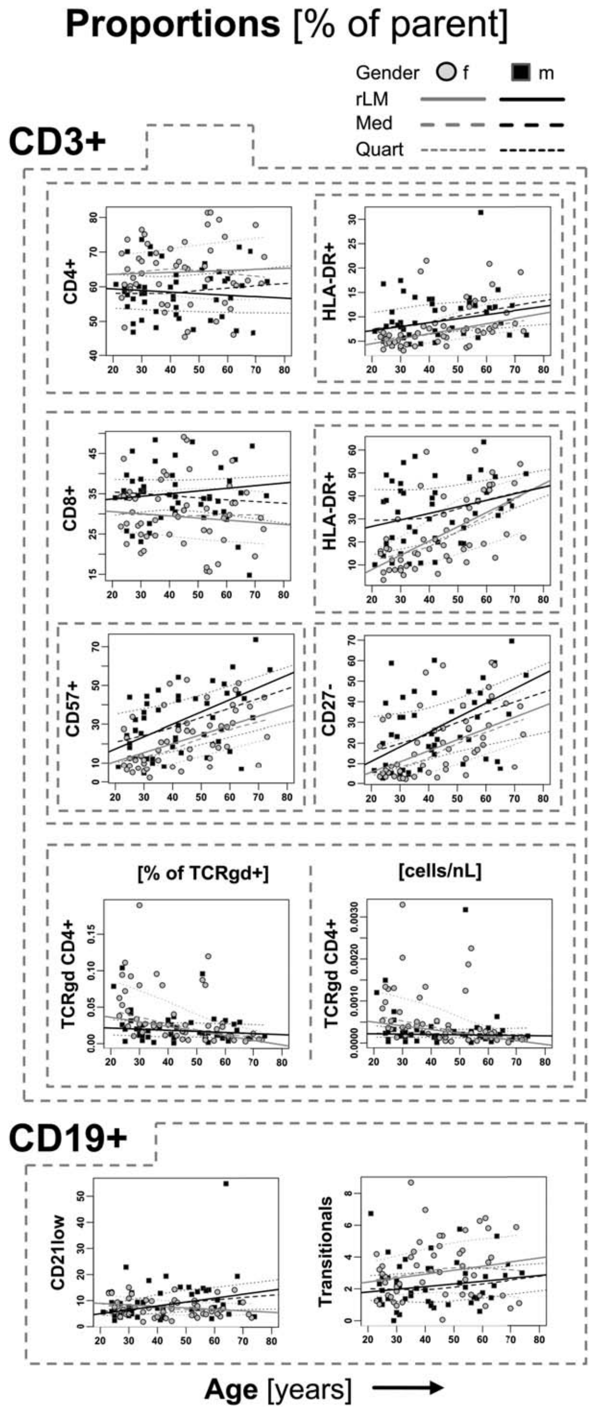

Figure 5. Gender- and age-dependent differences in lymphocytes. For selected lymphocyte subsets, median curves (Med), interquartile range curves (Quart) and robust linear regression lines (rLM) are plotted over age according to gender in proportional data of combined cohorts. Grouped subsets are encircled by dashed boxes. For the individual subset definitions see Table 2. *P*-values for cell populations, which significantly differ between females and males, are listed in Table 5.

Concomitantly they lose expression of the TNF family receptor CD27 and coreceptor CD28, while gaining CD57 and HLA-DR expression (53). In $CD8^+$ T cells the loss of CD28 and subsequently CD27 happens through repetitive cell replication and is considered a replicative end-stage where the cells acquire CD57 (20,58,59). In all regards, $CD8^+$ T cells showed an augmented differentiation compared to $CD4^+$ T cells, consistent with findings from several groups (22,53,60).

The same sequence of differentiation is not proven for $CD4^+$ T cells. Our data show significant increases of $CD4^+$ $CD27^-$ and $CD4^+$ $CD57^+$ T cells, while $CD4^+$ $CD28^-$ T cells remain constant. At the same time the $CD4^+$ $CD28^-$ subset is far smaller than the $CD4^+$ $CD27^-$ and $CD4^+$ $CD57^+$ subset, considering that the opposite is true for the $CD8^+$ compartment.

Differentiated T cells have a less diverse antigen receptor repertoire (60,61). Throughout life accumulation of differentiated T cells with similar antigenic specificities exhaust the pool of responsive T cells and leads to a progressive incompetence of the adaptive immune system, especially to $CD8^+$ T cell-dependent cytotoxic immune responses. The accumulation might be augmented from latent infections such as CMV, HSV, or HIV as several groups have shown (62–64). As illustrated in Figure 2, we also observed a considerable number of individuals with a high number of $CD4^+$ $CD57^+$ and $CD4^+$ $CD27^-$ T cells that could not be ascribed to aging alone, in contrast to $CD8^+$ T cells where only a proportion (<30%) of healthy individuals display a steady increase in their proportions during aging. The accumulation of differentiated $CD4^+$ T cells has been observed previously and may also result from latent viral infections such as described for their $CD8^+$ counterpart (65,66), but may also be caused by Th2 driven responses in unreported allergies or other subclinical conditions, which was not examined among our participants.

Our results reveal an augmented aging-associated T cell differentiation in males as compared with females (Fig. 5). This has been described previously (4,55) and may result from males being increasingly dominated by $CD8^+$ T cells. The $CD28^-$ T cell subset and the $CD4^+$:$CD8^+$ ratio are of particular interest because they both have been linked to mortality rates in the elderly (55). A male propensity for a fast aging $CD8^+$ T cell compartment may be a benefit in some immunological settings, that is, autoimmunity, but it may also lead to augmented immunosenescence, and could possibly be a contributor to the well-known reduced life expectancy in males.

We found no age or gender dependent correlation with Treg numbers. Previous studies have found both increasing and stable Treg numbers with age (22,67,68) and a recent study of 608 participants reported decreasing Treg numbers (56). Our data show that while the Treg compartment remains relatively stable, it undergoes an age-dependent differentiation toward a $CD45RA^-$ memory phenotype, not unlike other $CD4^+$ and $CD8^+$ T cells. Other studies have shown that CD45RA expression loss by Tregs is not accompanied by a decrease in suppressive capability (68,69), which points toward overall retained Treg function with age. One could still speculate that loss of naïve Tregs is accompanied by a loss of

Treg-TCR diversity, as it was shown for conventional T cells; however, studies remain to be performed. A study has linked the number of naïve Tregs to multiple sclerosis and it is possible that their disappearance interferes with self-tolerance (70).

We could not reproduce the total Treg number between our two cohorts. This particular subset was stained using two different protocols. In the first cohort, Tregs were stained according to the protocol described in the methods section, while a different staining protocol, that is, together with intracellular FoxP3, was applied in the second cohort. This methodical difference is likely to explain the inconsistency. Irreproducible numbers between protocols underlines the challenge of comparing identical cell subsets across different protocols and calls for standardization and strict adherence to operating procedures.

The $\gamma\delta$-TCR T cell compartment was dominated by $\gamma\delta$-TCR $CD4^{-}CD8^{-}$ T cells. $\gamma\delta$-TCR $CD4^{+}$ T cells showed a significant age-dependent decline. This association was caused by a steep age-dependent decrease in cell numbers from females. This subset is largely unknown but has previously been linked with IgE production in mice (71). A recent study reported the same age-dependent decrease in $\gamma\delta$-TCR $CD4^{+}$ T cells, but did not look at gender-specific differences (22). Our data show a possible female dominance of $\gamma\delta$-TCR $CD4^{+}$ T cells that might level out with advancing age. Subsets of such small sizes are very difficult to investigate and although confirmed in our second cohort, even larger studies are needed to verify our observations.

The total B cell compartment made up to 11.3% of all lymphocytes and showed no age- or gender-dependent relationship. The number of naïve $CD27^{-}$ and memory $CD27^{+}$ B cells also did not change with age. This finding is consistent with some reported findings (22,72), while others described significant age-dependent relationships (14,15). MZ B cells were the only B cell subset to show a marked age-dependent correlation (rho<-3). Though not significant after *P* values adjustment, our results suggest a decrease of MZ B cells with age. MZ B cells are not established until around the age of two years and they represent a fast responding B cell subset with a high prevalence in the MZ of the spleen (73). MZ B cells are important players against infection with polysaccharide microorganisms such as *Streptococcus pneumonia*, to which the elderly are particularly susceptible, and a diminished function or number of MZ B cells could contribute to their vulnerability (74,75).

In addition to revealing age- and gender-dependent differences in leucocyte subset distribution, we aimed to test the robustness and validity of reference values acquired by the recently established "ONE Study" flow cytometric panels. This was achieved by testing a second cohort of similar age and gender distribution. Given the relative moderate cohort sizes, we found remarkable reproducibility. The abundance of only a few leucocyte subsets could not be validated in our second cohort. This applied to absolute numbers of relatively low frequency populations, which proves that these rare cell types, especially when the cell has low or modulated marker expression, are the most difficult to standardize (33). Nonetheless, provided a sufficiently high biological variability, we demonstrate reproducible detection of age- and gender-specific differences at blood count levels around 1 cell per µl by standardized "ONE Study" flow cytometric profiling as in the case for $\gamma\delta$-TCR $CD4^{+}$ T cells. Here, a high staining index of all defining markers (CD3, CD4, and $\gamma\delta$-TCR) allows detection of few events with a precision that depends on Poisson counting statistics, that is, 11 or 25 events counted for a gated subset would have a high coefficient of variation of 30% or 20%, respectively. This can be considered an acceptable range when the biological variation is estimated to exceed the Poisson standard deviations several fold. Given that rare-event data are non-normally distributed, we have applied nonparametric as well as robust statistical testing procedures to assess age- and gender-specific differences with sufficient power. It is however important to conclude, that it would be absolutely necessary to increase the total number of acquired events when studying differences in such small subsets to get more statistical confidence, especially when analyzing individual samples or smaller cohorts.

Reproducibility and reference values are vital when patient variability is assessed, especially for multi-centre clinical trials applying flow cytometry. Our study and others have now demonstrated that the immune system is characterized by marked inter-individual and age-/gender variation that should be accounted for when interpreting clinical data. Here we report a standardized method with inter-operator reproducible results that can serve as a healthy reference group for future clinical trials with similar demographics. If the results are applicable to other demographics or still valid in a less controlled environment have yet to be established.

Conclusion

Several significant age dependent changes were observed through the age span of adulthood, which was the objective of our study. The age-related expansion of several senescent and pro-inflammatory cell subsets is consistent with theories of inflammaging and immunosenescence. The effects of age were most prominent to the adaptive immune system; in particular, $CD8^{+}$ T cells showed an accumulation of memory and senescent cells and concomitant loss of naïve cells, while the cell subsets of the innate immune system are much more conserved. Age proved to be a greater defining factor than gender for immunological heterogeneity. Several gender-related differences, such as $CD8^{+}$ dominance in male, may nonetheless be important to gender-specific diseases and mortality. Importantly, it is necessary that such age- and gender-related variability may be accounted for when interpreting immune monitoring results in clinical studies. Here, we have taken steps toward a sufficient standardization that allows comparison of data from different cohorts, and we have generated standardized and reproducible reference values.

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